Brian Wei

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Research Statement

My research develops **generative modeling** and **multimodal representation learning** methods for computer vision in **geospatial AI and ecology**. My work includes generative models for satellite imagery [1, 2, 3, 5], location representations of the Earth [4, 6, 7, 11], and vision models for biodiversity monitoring [7, 8, 9].

Education

Washington University in St. Louis — St. Louis, MO
Ph.D. in Computer Science — *Aug 2026 – Expected May 2030*
B.A. in Data Science | Minor in Economics; ***GPA: 3.99/4.00*** — *Aug 2022 – May 2026*

Vrije Universiteit Amsterdam — Amsterdam, Netherlands
IES Semester Abroad; ***GPA: 4.00/4.00*** — *Jan 2025 – May 2025*

Selected Publications

[1] **Wei, B.***, Sastry, S.*, Cher, D.*, Xing, E., & Jacobs, N. *TerraDiT-Ω: Unified Spatial Control for Satellite Image Synthesis with Any Geospatial Primitive.* In: European Conference on Computer Vision 2026 **(ECCV'26)** [paper] [code]

[2] Cher, D.*, **Wei, B.***, Sastry, S., & Jacobs, N. *VectorSynth: Fine-Grained Satellite Image Synthesis with Structured Semantics.* The IEEE/CVF Winter Conference on Applications of Computer Vision 2026 **(WACV'26)** [paper] [code]

[3] Sastry, S.*, Cher, D.*, **Wei, B.***, Dhakal, A., Khanal, S., Gupta, D., & Jacobs, N. *TerraDiT: Point-Conditioned Diffusion Transformer for Satellite Image Synthesis.* In: TerraBytes II Workshop, European Conference on Computer Vision 2026 **(ECCVW'26, Oral, Best Paper Award)** [paper] [code]

[4] Cher, D., Iqbal, H., Xing, E., **Wei, B.**, & Jacobs, N. *Tesselating The Earth.* In: European Conference on Computer Vision 2026 **(ECCV'26)** [paper] [code]

[5] Khanal, S., Cui, Y., Cher, D., Xing, E., **Wei, B.**, Sastry, S., & Jacobs, N. *Genesis: A Generative Engine for Hierarchical Satellite Image Synthesis.* The 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems **(SIGSPATIAL'26, Oral)** [paper] [code]

In Submission

[6] **Wei, B.**, Hsu, C., Gao, T., Cher, D., Xing, E., Beaulieu, D. & Jacobs, N. *X Marks the Spot: Diffusion for Image Geolocalization with Location Representations.* (under review in **ICLR'27**)

[7] Cher, D., **Wei, B.**, Xing, E., Jarosz, P., Sandvig, R., & Jacobs, N. *PhenoMap: Phenological Neural Fields for Ecological Mapping.* (under review in **ICLR'27**)

[8] Xing, E., **Wei, B.**, Cher, D., & Jacobs, N. *Learning to Allocate Tokens for Vision Transformers.* (under review in **ICLR'27**)

[9] Sastry, S., Cher, D., **Wei, B.**, Khanal, S., Gao, T., Dhakal, A., Xing, E., & Jacobs, N. *TaxaWalk: Species Recognition as a Walk Down the Tree of Life.* (under review in **ICLR'27**)

[10] Gao, T., Beaulieu, D., Hsu, C., **Wei, B.**, Zhang, X., Sastry, S., & Jacobs, N. *AeroEVA: Unified World Modeling and VLM Policy Learning for Embodied Aerial Search.* (under review in **ICLR'27**)

[11] Cher, D., Xing, E., Li, K., **Wei, B.**, Corley, I., & Jacobs, N. *Chronosphere: Space-Time Tessellation of Local Climate Experts.* (under review in **AAAI'27**) [paper]

Fellowships & Awards

NSF AI Advancements and Convergence in Computational, Environmental and Social Sciences (AI-ACCESS) National Research Traineeship (NRT) Fellowship (Aug 2026 – Aug 2028)

NSF Research Experiences for Undergraduates (REU) Fellowship. (May 2025 – Aug 2025)

WashU Statistics & Data Science (SDS) Summer Research Fellowship. (May 2024 – Aug 2024)

Research Experience

Multimodal Vision Research Laboratory St. Louis, MO

Graduate Research Assistant *Aug 2026 – current*

- Developing diffusion models for image geolocalization with location representations [6], hierarchical species recognition [9], and world models for embodied aerial search [10].
- Mentoring undergraduate students on ecology-focused computer vision projects.

Undergraduate Research Assistant *Sept 2025 – May 2026*

- Led development of the TerraDiT model suite, diffusion foundation models for satellite image synthesis with state-of-the-art spatial control from polygons, polylines, boxes, and points [1, 3].
- Worked on implicit neural representations of the Earth for phenology, climate, and general-purpose location encoding [4, 7, 11] and task-adaptive token allocation for vision transformers [8].

NSF REU Fellow *May 2025 – Aug 2025*

- Introduced VectorSynth, a diffusion model for satellite image synthesis spatially controlled by semantic maps [2].

SDS Summer Research Program St. Louis, MO

Undergraduate Research Fellow *May 2024 – Nov 2024*

- Designed a distributed spatial autoregressive model linking human mobility to mental health outcomes.

Teaching & Additional Experience

Washington University in St. Louis St. Louis, MO

Teaching Assistant *Jan 2026 – May 2026*

- Prepared and graded homeworks and quizzes for CSE 5509: Computer Vision.

WashU Robotics Club St. Louis, MO

Co-President & Software Lead *May 2023 – Aug 2024*

- Led a 100+ member organization and secured $25,000+ in engineering and student government grants.
- Led a 12-person team building autonomous waypoint navigation and teleoperation for a six-wheel rover in ROS2.

Engineering Test Kitchen St. Louis, MO

Machine Learning Engineer *Sept 2024 – Dec 2024*

- Built a CNN and data pipeline to detect and count staves in barrel production images.

Skills

Programming: Python, R, SQL, Java, Stata, XML.

AI/ML: Deep learning with PyTorch, Hugging Face, scikit-learn, PyTorch Lightning, diffusers, timm, TensorFlow/Keras, multi-GPU training.

Data/Visualization: pandas, NumPy, GeoPandas, SciPy, Tableau, matplotlib, Airflow, Snowflake.

Tools: Git, Linux/Bash, Docker, LSF/Slurm, LaTeX.